Ruben Baecker

Doctoral Student · Chair of Applied Cryptography · FAU Erlangen-Nürnberg

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Education

since 12/2023 — **Friedrich-Alexander-Universität Erlangen-Nürnberg**
Main focus: Threshold Cryptography, Adaptive Security, Oblivious Pseudorandom Functions, Password-Based Cryptography
Doctoral Student under Dominique Schröder, Cryptography

10/2021–09/2023 — **Friedrich-Alexander-Universität Erlangen-Nürnberg**
Main focus: Cryptography, Cyber Security, and IT-Forensics
Master of Science with Distinction, Computer Science
Thesis: Phoenix rises once again: How to defeat the (PW-)Hero
Completed while caring for a child full-time (02/2022–11/2023)

10/2017–05/2022 — **Friedrich-Alexander-Universität Erlangen-Nürnberg**
Main focus: Cryptography
Bachelor of Science with Distinction, Computer Science
Thesis: Securing Passwords against Offline Dictionary Attacks

Peer-reviewed

2026 — **Fully Adaptive FROST with Identifiable Aborts from AOMDL**
Ruben Baecker, Paul Gerhart, Davide Li Calsi, Luigi Russo, Dominique Schröder, and Arkady Yerukhimovich
ASIACRYPT

2026 — **Fair Distributed Exchange via Threshold Adaptor Signatures**
Ruben Baecker, Paul Gerhart, Jonathan Katz, and Dominique Schröder
ASIACRYPT

2025 — **Password-Hardened Encryption Revisited**
Ruben Baecker, Paul Gerhart, and Dominique Schröder
ASIACRYPT

2025 — **Universally Composable Password-Hardened Encryption**
Behzad Abdolmaleki, Ruben Baecker, Paul Gerhart, Mike Graf, Mojtaba Khalili, Daniel Rausch, and Dominique Schröder
ASIACRYPT

2025 **A Fully-Adaptive Threshold Partially-Oblivious PRF**
Ruben Baecker, Paul Gerhart, Daniel Rausch, and Dominique Schröder
CRYPTO

Preprints

2026 **Adaptive Distributed Key Generation for Discrete-Log Cryptosystems**
Ruben Baecker, Paul Gerhart, Stanislaw Jarecki, Phillip Nazarian, Daniel Rausch, and Dominique Schröder
eprint.iacr.org

Talks

2025 **Password-Hardened Encryption Revisited**
ASIACRYPT, Melbourne, AU
Slides

2025 **Universally Composable Password-Hardened Encryption**
ASIACRYPT, Melbourne, AU
Slides

2025 **A Fully-Adaptive Threshold Partially-Oblivious PRF**
CRYPTO, Santa Barbara, CA
Slides · Video

Work Experience

since 2023 **University Research Assistant**
Applied Cryptography
Friedrich-Alexander-Universität Erlangen-Nürnberg

2020–2021 **Student Employee**
Software Development in C#
Siemens AG - Digital Industries

2019–2020 **University Teaching Assistant**
System Programming I & II
Friedrich-Alexander-Universität Erlangen-Nürnberg

2018 **Student Employee**

Software Development in C#
Siemens AG - Energy Management

Scholarships and Awards

2024 **Jury Award at TechFak Science Slam**
500€
Awarded for the slam ”Password-Hardening - The Solution to the Password Problem” (in German)

2024 **ASQF Best Thesis Award**
500€
Awarded for the master thesis ”Phoenix Rises Once Again: How to Defeat the (PW-)Hero”

2020–2023 **German Academic Scholarship Foundation**
300€ per Month
Awarded for outstanding academic potential (extended in 2021 and 2022)

Parental Leave

12/2025–11/2026 **Part-time parental leave, working 30 h/week**
Friedrich-Alexander-Universität Erlangen-Nürnberg

05/2025–07/2025 **Full-time parental leave**
Friedrich-Alexander-Universität Erlangen-Nürnberg

08/2024–09/2024 **Full-time parental leave**
Friedrich-Alexander-Universität Erlangen-Nürnberg

Research Visits

08/2025–09/2025 **UC Irvine**
Hosted by Prof. Stanislaw Jarecki
USA

Supervised Students

2026 **Samuel Huber**
An Analysis of SecureDNA Continued
Master Thesis

2025 **Tabea Stengl**
An Analysis of SecureDNA
Bachelor Thesis

2025 **Matthias Flosbach**
Parameter Selection for LWR-based PRFs
Bachelor Thesis

Teaching Activities

Winter 26/27 **Introduction to Modern Cryptography**
Teaching Assistant

Summer 26 **Cryptography and Its Impact**
Organizer

Winter 25/26 **Cryptography in Secure Messaging**
Organizer

Summer 25 **Cryptography and Its Impact**
Organizer

Winter 24/25 **Cryptography in Secure Messaging**
Organizer

Summer 24 **Introduction to Algorithms (in German)**
Guest Lecturer

Community Service

Program Committee
27 USENIX
26 ESORICS

External Reviews (Selected)
24,25 CRYPTO
24,26,27 EUROCRYPT
26 ASIACRYPT
24,25 CCS
25 PKC
24,25,26 ESORICS
24,26,27 CSF